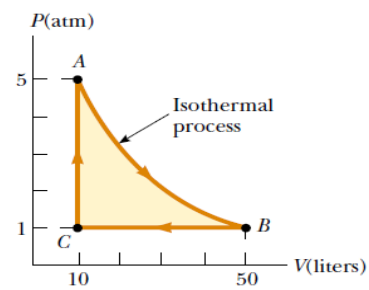
# PHY101: Introduction to Physics-1

***Monsoon, 2018***

## Tutorial-16

1. An ideal gas with the adiabatic exponent $\gamma$ goes through a direct (clockwise) cycle consisting of adiabatic, isobaric, and isochoric lines. Find the efficiency of the cycle if in the adiabatic process the volume of the ideal gas
(a) increases n-fold; (b) decreases n-fold.

2. Calculate at the temperature T = 17 °C;
(a) the root mean square velocity and the mean kinetic energy of an oxygen molecule in the process of translational motion.
(b) the root mean square velocity of a water droplet of diameter d = 0.10 μm suspended in the air.

3. A 1 mol sample of an ideal monatomic gas is taken through the cycle as shown in figure. The process A → B is a reversible isothermal expansion. Calculate (a) the net work done by the gas, (b) the energy added to the gas by heat, (c) the energy exhausted from the gas by heat, and (d) the efficiency of the cycle.